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When Graph Autoencoders Look the Wrong Way:

Complexity Bias and Simple Baselines in Unsupervised Anomaly Detection on the LHC Olympics 2020 Dataset

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Abstract

Graph autoencoders (GAEs) trained only on background are a natural candidate for model-agnostic new-physics searches. I tested two of them on the LHC Olympics 2020 R&D dataset, where a $W' \to XY \to (q\bar{q})(q\bar{q})$ dijet resonance is hidden in QCD background. The first is a node-level GAE that scores per-particle reconstruction error. The second encodes the whole event, reconstructs it with a set decoder and a Chamfer loss, and offers three scores: reconstruction error, latent k-nearest-neighbour distance and latent Gaussian-mixture likelihood. The decision criterion was fixed before looking at any test result. Neither model beats simple label-free observables. The node-level model reaches an AUC of 0.483 ± 0.005 with max-reduction and 0.781 with mean-reduction; all three event-level scores are *inverted*, with AUCs between 0.401 and 0.436, stable across seeds, nine ablations, and signal-contaminated training. By contrast, the mean transverse momentum per particle, $\langle p_{\mathrm{T}} \rangle = H_{\mathrm{T}}/N$, reaches 0.811, and $-\tau_{21}$ of the leading jet reaches 0.801. Correlation and binned analyses indicate that the latent-density scores track event complexity: the signal has fewer, harder particles than QCD and is therefore placed in dense regions of the learned manifold. The benchmark code is public, and I suggest $\langle p_{\mathrm{T}} \rangle$ as a standard reference baseline for unsupervised methods on this dataset.

Code and data: https://github.com/Viverun/Hidden-Vertex
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1 Introduction

Searches for physics beyond the Standard Model (BSM) at the Large Hadron Collider (LHC) are mostly designed around specific signal hypotheses. Model-agnostic anomaly detection aims to complement them by flagging events that are atypical with respect to the Standard Model background, without assuming a signal model. The LHC Olympics 2020 (LHCO) [1] established a common benchmark for this task, and autoencoders, in which events with large reconstruction error are treated as anomalous, are among the most widely studied unsupervised approaches [2–5].

Autoencoder scores have a known weakness: they can end up measuring how complex an input is rather than how unusual it is. For jet images, simple signal jets turn out to be *easier* to reconstruct than QCD jets [4]. This project started as an attempt to build a working graph-based detector, and ran into the same problem for particle graphs of whole events. This paper documents what happened:

1. a reproducible benchmark harness on the public LHCO R&D data, with a frozen test set, multi-seed runs, bootstrap confidence intervals and within-bin AUCs that test whether a score carries information beyond trivial event quantities;

2. a set of label-free reference baselines, among which the mean transverse momentum per particle, $\langle p_{\mathrm{T}} \rangle = H_{\mathrm{T}}/N$, is the strongest particle-level observable (AUC 0.811), above H_{T}, m_{jj}, and a node-level GAE;

3. a negative result, under a decision rule fixed in advance, for an event-level GAE whose three anomaly scores are systematically inverted, with ablations, a contamination study, and a correlation-based diagnosis that links the latent-density inversion to event complexity.

None of this is a working anomaly detector. The practical point is simpler: before claiming that an unsupervised model has learned something on this dataset, compare it with $\langle p_{\mathrm{T}} \rangle$. If it cannot beat a ratio of two event totals, it has probably only learned the energy and multiplicity scale.

2 Dataset and test protocol

I use the LHCO 2020 R&D dataset [1, 6]. Background consists of QCD dijet events and signal of $W' \to XY$ with $m_{W'} = 3.5$ TeV, $m_X = 500$ GeV and $m_Y = 100$ GeV, each of X and Y decaying to two quarks. Events were generated with Pythia 8 [7] and the Delphes 3 [8] detector simulation, and are stored as up to 700 hadrons $(p_{\mathrm{T}}, \eta, \phi)$. Jet-level features (anti-k_t [9] $R = 1$ jet masses, N-subjettiness ratios [10] and m_{jj}) are taken from the accompanying feature file.

One practical warning first: the publicly available file `events_anomalydetection_v2.h5` is *sorted*: rows 0–999,999 are background and rows $\geq$ 1,000,000 are signal. Any pipeline that reads a prefix of the file therefore sees no signal at all; I select events explicitly by row index. The test set is frozen across all experiments: 5,000 background events (rows 40,000–44,999) and 5,000 signal events (rows 1,000,000–1,004,999). Models are trained on 180,000 background-only events (rows $\geq$ 45,000) with a 20,000-event background validation set.

Metrics. I report the ROC AUC with bootstrap confidence intervals over the 10,000 test events (95% for baselines; 98.3% for the three model scores, a Bonferroni correction for three comparisons). To test whether a score carries information beyond trivial event quantities, I also compute the AUC *within bins* of H_{T}, particle multiplicity N, and leading-particle p_{T}. The sign of every score ("larger = more anomalous") is fixed in advance.

Table 1: ROC AUC of label-free event observables and the node-level GAE on the frozen test set (5k signal, 5k background), with 95% bootstrap intervals and the minimum AUC within bins of H_{T}, N and leading p_{T}. Node-level GAE rows are from seed 0; over five seeds the max-reduction AUC is 0.483 ± 0.005.

Score	Type	AUC	95% CI	min. within-bin
$-\lvert m_{jj} - 3500\,\mathrm{GeV}\rvert$	oracle	0.900	0.894–0.906	0.777
$\langle p_{\mathrm{T}} \rangle = H_{\mathrm{T}}/N$	particle	**0.811**	0.803–0.821	0.599
$-\tau_{21}$ (leading jet)	jet	0.801	0.792–0.810	0.664
$H_{\mathrm{T}} = \sum p_{\mathrm{T}}$	particle	0.792	0.783–0.802	0.337
m_{jj}	jet	0.788	0.778–0.799	0.322
Node-level GAE (mean)	model	0.781	0.772–0.790	0.609
$-N$	particle	0.715	0.704–0.726	0.399
m_{j2} / m_{j1}	jet	0.637 / 0.634	–	0.566 / 0.541
leading-particle p_{T}	particle	0.554	0.543–0.565	0.388
Node-level GAE (max)	model	0.478	0.467–0.489	0.376

Decision criterion. Before any test result for the event-level model, I fixed the following gate: a model score passes only if the lower bound of its seed-averaged AUC interval exceeds the best particle-level baseline, and its minimum within-bin AUC is at least 0.6.

3 Models

Node-level GAE. Each event is a radius graph with edges between particles at $\Delta R < 0.4$. A graph convolutional encoder [11] compresses each particle's features to a one-dimensional latent and a decoder reconstructs its standardized $(p_{\mathrm{T}}, \eta, \phi)$; the model has 420 effective parameters. The event score is the maximum or the mean of the per-particle reconstruction errors.

Event-level GAE. The encoder applies three residual neighbour-averaging layers of width 64 on the same $\Delta R < 0.4$ graph, followed by concatenated mean, max and p_{T}-weighted pooling into a 16-dimensional event latent. A set decoder reconstructs the $K = 48$ hardest particles and is trained with a symmetric Chamfer distance [12], which is invariant to particle ordering. Events are truncated to the 350 hardest particles, a cut chosen from the measured multiplicity distribution before training (0.7% of background and 0.02% of signal events exceed it). The model has 95,888 parameters and is trained for 30 epochs with AdamW (learning rate 10^{-3}, batch size 256, early-stopping patience 5) on one NVIDIA T4 GPU, about 18 minutes per seed. Three anomaly scores are evaluated: (i) the Chamfer reconstruction error; (ii) the mean distance to the 20 nearest training latents; and (iii) the negative log-likelihood of the latent under an 8-component Gaussian mixture fitted to training latents.

4 Results

4.1 Reference baselines

Table 1 lists label-free event observables evaluated on the frozen test set. The harness reproduces my earlier node-level numbers exactly (H_{T} 0.792, m_{jj} 0.788, node-level GAE 0.478/0.781, each inside its interval). The strongest particle-level observable is $\langle p_{\mathrm{T}} \rangle$ (AUC 0.811), which exceeds H_{T} because it combines the higher energy scale of signal with its lower multiplicity (mean $N = 140$ for signal vs. 182 for background). The mass-window score $-\lvert m_{jj} - 3500\,\mathrm{GeV}\rvert$ (AUC 0.900) requires knowledge of the signal and is shown only as an oracle reference.

Table 2: Event-level GAE: seed-averaged AUC over three seeds with 98.3% bootstrap intervals, minimum within-bin AUC, and per-seed values.

Score	AUC	98.3% CI	min. within-bin	seeds 0 / 1 / 2
Chamfer reconstruction	0.428	0.415–0.442	0.311	0.419 / 0.433 / 0.432
Latent kNN distance	0.401	0.389–0.412	0.281	0.405 / 0.397 / 0.401
Latent GMM likelihood	0.436	0.426–0.448	0.308	0.448 / 0.427 / 0.434

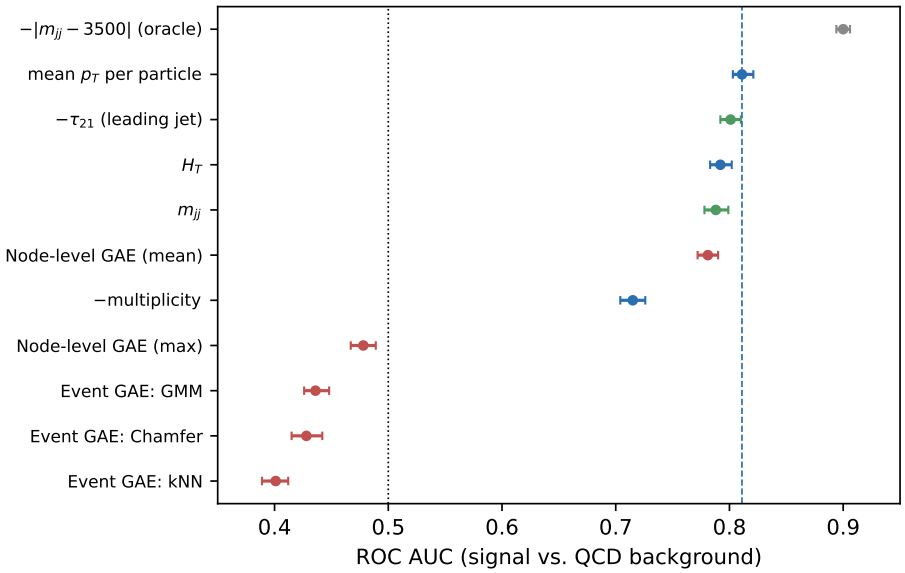


Figure 1: ROC AUC with confidence intervals for reference observables (blue: particle level; green: jet level; grey: oracle) and autoencoder scores (red). The dotted line marks chance and the dashed line the $\langle p_{\mathrm{T}} \rangle$ baseline.

For the node-level GAE, the choice of reduction matters more than the model: the same trained network gives 0.48 with max and 0.78 with mean. Across five seeds, the max-reduction AUC is 0.483 ± 0.005. Scaling the network by 44× in parameters lowers the training loss by about 2% and leaves the AUC unchanged. The max score is dominated by the hardest particle (Spearman $\rho = 0.70$ with leading p_{T}, and AUC 0.38–0.47 within leading-p_{T} deciles). Since a per-particle MLP with the same one-dimensional bottleneck reconstructs p_{T} almost perfectly, the residual error originates in the neighbourhood averaging of the graph convolution. The mean score retains AUC 0.61–0.82 within deciles of H_{T}, N and leading p_{T}, but it is strongly anti-correlated with multiplicity ($\rho = -0.71$) and does not exceed $\langle p_{\mathrm{T}} \rangle$.

4.2 Event-level GAE

Table 2 summarises the headline result. All three scores lie significantly *below* 0.5, and the gate is not passed. The seed-to-seed spread (0.01–0.02) is small compared with the distance to 0.5, so the inversion is systematic. Training was stable, with agreement between training and validation loss and no latent collapse.

Figure 1 compares all scores. Every model score lies below the simplest particle-level observables.

Table 3: Sensitivity of the event-level GAE (seed-averaged AUC). Main configuration: latent 16, $K = 48$, depth 3, full pooling, unweighted Chamfer loss. Contamination rows use one seed.

Variant	seeds	Chamfer	kNN	GMM
main	3	0.428	0.401	0.436
latent 4	2	0.366	0.405	0.403
latent 8	2	0.401	0.372	0.376
latent 32	2	0.428	0.412	0.485
$K = 24$	2	0.502	0.447	0.475
$K = 96$	2	0.490	0.425	0.447
depth 2	2	0.421	0.412	0.442
depth 4	2	0.429	0.388	0.428
no p_{T}-weighted pooling	2	0.424	0.408	0.451
p_{T}-weighted Chamfer loss	2	0.475	0.454	0.468
pure background (seed 0)	1	0.419	0.405	0.448
1% signal in training	1	0.424	0.420	0.461
9% signal in training	1	0.413	0.362	0.387

4.3 Ablations and contamination

Nine variations of the event-level model, each with two seeds, were run as a sensitivity analysis and not used to select a configuration (Table 3). None produces a useful score; the best single value is 0.502 (Chamfer, $K = 24$), i.e. chance. Training on background contaminated with 1% signal changes the scores by at most ~ 0.015, which is inconclusive with one seed. At 9% contamination, the signal fraction of the R&D file, the density scores become *more* inverted (kNN 0.362, GMM 0.387), consistent with signal events making signal-like regions of the latent space denser.

5 Diagnosis

Table 4 gives Spearman correlations of the seed-0 scores with simple event quantities. The latent-density scores increase with multiplicity ($\rho = +0.28$ for kNN, $+0.23$ for GMM) and decrease with $\langle p_{\mathrm{T}} \rangle$ (-0.27, -0.20), also within background alone. Their AUC within multiplicity bins is close to chance (kNN 0.43–0.50, GMM 0.47–0.51). Since signal events have fewer and harder particles than QCD background, they fall into dense regions of a latent space trained on QCD. Under the pre-declared sign (far from training data means anomalous), this produces AUCs below 0.5. This explanation is supported by correlations but was not tested causally.

The Chamfer score behaves differently: it barely tracks multiplicity ($\rho = +0.08$) or $\langle p_{\mathrm{T}} \rangle$ ($\rho = -0.07$), and remains inverted within multiplicity bins (0.41–0.48) and within H_{T} bins (down to 0.31). Its inversion is therefore not explained by multiplicity alone, and its origin remains open. The three scores are only moderately correlated with one another (Chamfer–kNN 0.53, Chamfer–GMM 0.42, kNN–GMM 0.87). Flipping the sign post hoc would give AUCs of 0.56–0.60, still far below $\langle p_{\mathrm{T}} \rangle$. I mention this only to show that the scores contain a weak complexity signal rather than noise; it is not a result.

6 Discussion and outlook

These results add to the evidence that autoencoder-based anomaly scores are vulnerable to complexity bias [4], and extend it to particle-graph representations of full events. Two practical lessons follow. First, unsupervised methods on the LHCO R&D data should be compared not

Table 4: Spearman correlation of seed-0 event-level scores with event quantities (all test events / background only / signal only).

Score	H_{T}	N	leading p_{T}	$\langle p_{\mathrm{T}} \rangle$
Chamfer	-0.01 / $+0.15$ / 0.00	$+0.08$ / $+0.08$ / -0.01	$+0.17$ / $+0.18$ / $+0.19$	-0.07 / -0.02 / $+0.01$
kNN	-0.02 / $+0.19$ / 0.00	$+0.28$ / $+0.30$ / $+0.19$	-0.02 / -0.05 / $+0.05$	-0.27 / -0.24 / -0.20
GMM	$+0.02$ / $+0.18$ / 0.00	$+0.23$ / $+0.27$ / $+0.15$	$+0.01$ / -0.04 / $+0.09$	-0.20 / -0.21 / -0.16

only with H_{T} but with $\langle p_{\mathrm{T}} \rangle$, which already reaches AUC 0.81 without any training. Second, within-bin AUCs at fixed multiplicity and H_{T} are a cheap and informative diagnostic: a score that loses its discrimination at fixed complexity has not learned substructure.

The features that distinguish the signal in this dataset are jet-level: two-prong substructure (τ_{21}), jet masses and m_{jj}. These are not available to the encoder as explicit inputs. Natural next steps are therefore (i) adding jet-level and substructure inputs, (ii) conditioning the model on, or regressing out, multiplicity and $\langle p_{\mathrm{T}} \rangle$ so that the score measures structure at fixed complexity, and (iii) choosing the score sign on background-only or simulated data. Large foundation models for collider physics trained on simulated events are expected to face the same issue. Benchmarks of the kind released here can be used to check whether their anomaly scores capture more than event complexity.

Limitations. All results use a single signal model and the R&D dataset, not the LHCO black boxes. The event-level model was evaluated in one configuration fixed in advance with limited ablations, and the complexity explanation is correlational. Ablation and contamination runs use two or one seeds, so differences below ~ 0.02 should not be interpreted.

Code availability

All code, configurations, and the benchmark harness are available at the public repository `https://github.com/Viverun/Hidden-Vertex`, under the MIT licence. The detailed result reports are in `docs/PART1_VERIFICATION.md` and `docs/PART2_RESULTS.md`.

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Use of AI tools

AI assistants (Anthropic's Claude) were used to help with code development, debugging and editing of the manuscript text. All experiments, results and conclusions were checked by the author, who takes full responsibility for the content of this paper.

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